

**Species Diversification in the Sky Islands of Southwestern China Revealed
by Genomic, Introgression and Demographic Analyses of Asian Shrew
Moles**

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Abstract

The Mountains of Southwest China, a global biodiversity hotspot, have a unique “sky island” landscape with high diversity of both ancient and recent-formed species. While their distribution patterns offer significant insights into diversification processes, the complex geological and climatic history, combined with dynamic histories of gene flow in endemic taxa, make unravelling this history challenging. This study focuses on Asian shrew moles (genus *Uropsilus*), an ancient group endemic to this region with an unresolved taxonomic system. By combining phylogenomic, introgression and demographic history analyses, we investigated the historical patterns of species diversification in this genus. We detected phylogenetic discordances among rapidly diverged lineages, driven by incomplete lineage sorting, both recent and ancient gene flow, and ghost introgression. The gene flow patterns revealed strong genetic isolation in the Hengduan Mountains region, contrasted by more extensive dispersal or connectivity in areas to its east, while suggesting potential ring-like diversification around the Sichuan Basin. Demographic history indicated that rapidly diverged lineages south of the Yangtze River exhibited significantly different responses to climatic fluctuations compared to other lineages, with the East Asian monsoon likely driving their radiative differentiation and dispersal. Our study demonstrates the impacts of mountain

uplift, climatic changes, and the connectivity of sky island refugia in shaping the diverse patterns of species differentiation and their distribution. [phylogenomics; introgression; Asian shrew moles; demographic history]

Mountain regions, accounting for only one-fifth of the world's land area, harbor half of its biodiversity hotspots (Myers et al. 2000; Marchese 2015). The formation and maintenance of biodiversity in mountainous areas is facilitated through various mechanisms (Rahbek et al. 2019a, 2019b; Perrigo et al. 2020): 1) Orogenic processes cause high environmental heterogeneity and drive the isolation of species ranges over geologically short time scales, promoting rapid species differentiation; 2) Elevational environmental gradients in mountainous areas allow ancient taxa to be preserved during rapid environmental changes; and 3) Continuous mountain ranges can serve as corridors for species migration and dispersal to occupy new ecological niches. The Mountains of Southwest China constitute one of the thirty-six global biodiversity hotspots, representing one of the most biodiverse regions in the Northern Hemisphere (Myers et al. 2000; Yang et al. 2004; Lei et al. 2015). The generally connected yet locally isolated complex of "sky islands" in this region, coupled with each island's provision of a stable environment, facilitates the formation, expansion and preservation of biodiversity (He and Jiang 2014). Vicariance has long been regarded as one of the main factors driving the diversification of species in the sky islands in the region (He and Jiang 2014; Wan et al. 2018; Dong et al. 2020). However, present species distribution patterns are often the result of the dynamic interplay of topography, climate, interspecific gene flow,

and the establishment of reproductive isolation (Pavón-Vázquez et al. 2024). This complex interplay poses significant challenges for understanding the diversification history of ancient lineages.

Asian shrew moles, *Uropsilus* spp., are one of the most representative ancient taxa in the Mountains of Southwest China. *Uropsilus* is the sole living genus of Uropsilinae and is sister to all other Talpidae (Douady and Douzery 2003; Sanchez-Villagra et al. 2006; He et al. 2016). It diverged from other talpid subfamilies at the beginning of the Eocene (Springer et al. 2018). The fossil record of Uropsilinae spans from the Miocene to the present across Europe, Asia, and North America (García-Alix et al. 2011; Kietmann et al. 2013; Qiu and Storch 2005; Ziegler 2012). All other Uropsilinae genera went extinct before the early Pleistocene, leaving *Uropsilus* endemic to the Mountains of Southwest China and adjacent mountains (MSCAM). *Uropsilus* has evolved into a highly diverse group comprising 11 recognized species and at least six cryptic species (Wan et al. 2018; Hu et al. 2021; Bui et al. 2023; Ren et al. 2023; Jiang et al. 2025). These small animals are widely distributed in the MSCAM, where they strongly prefer cool, humid mountain forests at mid-to-high-elevations, resulting in pronounced geographical structure across the sky island terrains. Consequently, long-term isolation and niche conservatism have been considered the primary drivers of their speciation (Wan et al. 2018). However, our understanding of this group's population dynamic history and interspecific gene flow remains limited, which may lead to overlooking or misinterpreting other key factors influencing species differentiation, such as climate fluctuations and topographic connectivity, and may further complicate taxonomic classification and species status assessments. Although the number of newly discovered

Uropsilus species has increased in recent years, phylogenetic trees constructed from limited gene fragments exhibit significant conflicts, leaving the status of multiple putative species unresolved (Wan et al. 2013, 2018; Hu et al. 2021; Bui et al. 2023; Ren et al. 2023; Jiang et al. 2025). For example, under the current taxonomic framework, *Uropsilus gracilis* constitutes a polyphyletic species complex comprising multiple cryptic species, whose evolutionary history remains unclear.

Exploring the evolutionary and diversification history of species requires a robust species tree. However, mito-nuclear incongruences, incomplete lineage sorting (ILS) and introgression pose significant challenges to constructing an accurate species tree (Degnan and Rosenberg 2009; Meyer et al. 2016). Approaches for phylogenetic heterogeneity analyses and introgression detection based on whole-genome data have increasingly improved. For instance, the D_{FOIL} statistic based on five-taxon phylogeny performs well in the presence of ILS and at low rates of introgression, and can infer ancient introgression signals (Pease and Hahn 2015; Meleshko et al. 2021). Some coalescent- and network-based approaches can reconstruct the phylogenetic network of species and suggest the presence of “ghost lineages” (unsampled lineages) (Solís-Lemus and Ané 2016; Zhang et al. 2024). These approaches not only aid in identifying an accurate species tree but also offer valuable insights into the complex processes of species differentiation.

In this study, we conducted comprehensive phylogenetic sampling and whole-genome resequencing of known and cryptic species of Asian shrew moles. We assessed how mito-nuclear incongruences, ILS, and gene flow contribute to phylogenetic conflicts and reconstructed a species tree using a coalescent-based method. Our analyses also detected

multiple ancient introgression events, including potential ghost introgression. Furthermore, we combined biogeographical and population dynamic history analyses to examine how topography and climate fluctuations have shaped species diversification histories and distribution patterns in the Mountains of Southwest China. Our results strongly support the species status of each putative species and underscore the need for future taxonomic clarification of the genus through integrative approaches.

MATERIALS AND METHODS

Sampling, Genomic Resequencing, and Variant Calling

We selected 39 samples for whole-genome resequencing from 16 species/cryptic species based on previous studies (Ren et al. 2023; Bui et al. 2023). We ensured that at least one sample was selected from each defined geographic unit across every species or cryptic species (Fig. 1a; Supplementary Fig. S1; Supplementary Table S1). These collections were approved by the Animal Care and Ethics Committee of the Kunming Institute of Zoology, Chinese Academy of Sciences (KIZ), and the wildlife management administrative offices in the areas where *Uropsilus* spp. were sampled. Tissue samples, including muscle and liver samples, were preserved in ethanol at -80°C before DNA extraction. The whole genome resequencing was performed on an Illumina NovaSeq 6000 platform. Details of library preparation and sequencing procedures are provided in the Supplementary Materials.

We employed fastp v0.19.7 (Chen et al. 2018) for basic statistics and quality control of raw reads (FastQ format). The trimmed reads were mapped to the *Uropsilus gracilis* reference genome (Liu et al. 2023) using BWA-MEM v0.7.17 (Li 2013) with default

parameters. Sequence Alignment/Map (SAM) files were imported into Picard v2.25.6 (<https://broadinstitute.github.io/picard/>) for sorting and removing duplicate PCR reads. Single-nucleotide polymorphism (SNP) calling, genotype calling and SNPs filtering were conducted via the Genome Analysis Toolkit (GATK v4.2.0.0) (McKenna et al. 2010); detailed procedures and parameters are provided in the Supplementary Methods. After filtering, we merged all VCF files and retained SNPs with a missing rate of less than 20% using VCFtools v0.1.16 (Danecek et al. 2011).

Phylogenomic Analyses

Phylogenetic relationships among species were inferred using multiple methods. First, we conducted maximum likelihood (ML) analyses with RAxML v8.2.12 (Stamatakis 2014). To optimize computational efficiency, the dataset, including both invariable sites and SNPs, was thinned by selecting one site for every 100 bp across the genome, and only concatenated SNPs were used to conduct ML analysis. To correct the bias introduced by using only variable sites (Leaché et al. 2015), we also recorded the number of invariable sites that were filtered out from the down-sampled dataset based on the script `ascbias.py` (available at https://github.com/btmartin721/raxml_ascbias). Heterozygous sites, represented by IUPAC ambiguity codes, were retained as variants, while only truly invariable sites (columns with a single nucleotide state) were removed for the ascertainment bias correction. In total, 1,437,195 SNPs were retained, and the corresponding number of invariable sites used for calibration was 13,567,706. We employed the nucleotide substitution model ASC_GTRGAMMA, and estimated bootstrap support values using a rapid bootstrapping

algorithm with 500 pseudoreplicates. Next, we utilized PLINK v1.90 (Chang et al. 2015) to obtain pairwise estimates of identity-by-state (IBS) scores for all samples and constructed a neighbor-joining (NJ) tree with Phylip v3.698 (Felsenstein 2009).

Due to the high degree of divergence between Uropsilinae and other Talpidae taxa, even the closest outgroup cannot be accurately mapped to the *Uropsilus* reference genome, leading to the unavailability of an outgroup in the SNPs dataset. Thus, we further constructed an ML tree using genome coding sequences (CDSs) with an outgroup to determine the placement of the root of the *Uropsilus* phylogeny. We used the *Talpa occidentalis* (Iberian mole) as an outgroup and downloaded its CDSs from NCBI GenBank database under accession number GCA_014898055.3. We selected one individual from each clade and used Gffread (Pertea and Pertea 2020) to obtain the CDSs of the *Uropsilus* species. Sequences with in-frame stop codons, not multiples of three in length, or translated to fewer than 50 amino acids were removed. A total of 5,815 single-copy genes were identified across the 16 *Uropsilus* lineages and outgroup *Talpa occidentalis* using Orthofinder v2.5.1 (Emms and Kelly 2019), and four-fold degenerate sites (CDS-4dtv) were extracted. A phylogenetic tree was constructed with CDS-4dtv dataset using IQ-TREE v2.1.4 (Minh et al. 2020), employing the “-m MFP” parameter (ModelFinder Plus (Kalyaanamoorthy et al. 2017)) to automatically conduct model selection.

For mitochondrial phylogenetic analysis, we assembled mitogenomes (deposited in GenBank under accession numbers: PV938352-PV938355; PV939677-PV939711) using MitoZ v2.3 (Meng et al. 2019) or GetOrganelle v1.7.2 (Jin et al. 2020) (see Supplementary Methods) and constructed an ML tree with a concatenated dataset comprising 13

protein-coding genes and two ribosomal RNA (rRNA) genes in RAxML v8.2.12. Before constructing the mitochondrial tree, we aligned each gene with ClustalW v2.1 (Larkin et al. 2007) and calculated partitioning schemes and best-fit model of evolution for each partition (gene) using PartitionFinder v2.1.1 (Lanfear et al. 2017).

In addition, we constructed a species tree using a coalescent-based method. We obtained the nuclear genome consensus sequences from BAM (Binary Alignment/Map) files for each individual using ANGSD v0.935 (Korneliussen et al. 2014). Then we divided them into approximately 40,000 windows, each 50 kb in length, and selected 10 kb for each window to reduce the length and obtain non-adjacent windows. Sequences with missing data > 20% were removed, which generated 33,606 alignments. We constructed a gene tree for each sequence using IQ-TREE v2.4.0, with substitution models automatically selected for each partition (i.e., each sequence) through the default ModelFinder Plus (-p MFP) algorithm. A species tree was inferred from these gene trees using ASTRAL v5.7.8 (Zhang et al. 2018), with each individual treated independently. To investigate the phylogenetic discordance across the genome, we estimated gene, site, and quartet concordance vectors (gCVs, sCVs and qCVs) for branches following an online tutorial using IQ-TREE and ASTRAL (<http://www.iqtree.org/doc/recipes/concordance-vector>) (Lanfear and Hahn 2024).

Divergence Time Estimation

Divergence times among *Uropsilus* species were estimated using two methods: SNAPP package of BEAST2 v2.6.6 (Bryant et al. 2012) and the PAML-MCMCTree program (Yang 2007). We employed two secondary calibration points. (1) Based on the earliest divergence

time of Talpidae estimated by Springer et al. (2018) (47.48 Ma, 95% CI = 35-57 Ma), we set the calibration point for crown Talpidae using a normal distribution with mean = 47.48, standard deviation = 6.08. For the MCMCTree analysis, we set the lower boundary at 35 Ma and the upper boundary at 57 Ma. (2) Based on the estimated earliest divergence time of *Uropsilus* from He et al. (2021) at approximately 8 Ma, we employed a normal distribution (mean = 8, standard deviation = 2.4). For the MCMCTree analysis, we set the lower boundary at 4 Ma and the upper boundary at 12 Ma.

We used the CDS-4dtv dataset because these sites conform better to the assumptions of neutral evolution. We selected one individual per clade. For SNAPP analysis, only biallelic SNP sites were used. To improve computational efficiency while maintaining analytical robustness, we thinned the dataset by retaining only one SNP per 100 SNPs, resulting in a final dataset of 4,196 SNPs. The SNAPP analysis was run for 10 million Markov Chain Monte Carlo (MCMC) iterations to co-estimate the species tree and divergence times, with the first 20% of iterations as burn-in. For the MCMCTree run, the ML tree constructed from the CDS-4dtv dataset was used as the input tree. We conducted 10 million MCMC iterations and discarded the first 20% of the iterations as burn-in. Tracer v1.7.2 (Rambaut et al. 2018) was used to assess the stationarity of the parameter estimates in both analyses and the convergence of the MCMC runs, with ESS values > 200 considered indicative of stationarity. TreeAnnotator v2.6.6 (Helfrich et al. 2018) was used to estimate a maximum clade credibility tree.

Phylogenetic Network and Introgression Tests

We employed SNaQ (Species Networks applying Quartets) method from the PhyloNetworks package (Solís-Lemus and Ané 2016) to investigate the introgression history among lineages. We selected one individual per clade and pruned SNPs by selecting one SNP per 1,000 bp window across the genome. We computed concordance factors from the SNPs datasets using an R function, SNPs2CF v1.6 (Olave and Meyer 2020). The table of concordance factors was used as input for SNaQ. The species tree topology was used as the starting tree. We assumed the maximum number of hybrid nodes (h_{max}) ranged from 0 to 10 for phylogenetic network inference. For each h_{max} value's analysis, 10 independent runs were conducted. The results with the lowest -loglik score (pseudo deviance) were considered the best inference.

For the introgression analysis, we removed scaffolds less than 1,000,000 bp from the reference genome FASTA file. We calculated the D statistic and f_4 admixture ratio and used the f-branch [or fb(C)] metric to disentangle correlated D or f_4 ratio results using Dsuite v0.4 r43 (Malinsky et al. 2021). Due to the absence of a suitable outgroup for the introgression analysis across the entire genus, we used *Uropsilus investigator* (Clades A and B) (Fig. 1b) as the outgroup, as its deep divergence and geographical isolation preclude genetic exchange with other species (Wan et al. 2018). We also performed an additional D test using Dquartets in Dsuite, which does not require an outgroup, to ensure the reliability of the results using *Uropsilus investigator* as the outgroup. In addition, we divided Clades I, M and P (see Fig. 1b) into two subclades (Clades I1, I2, M1, M2, P1, P2) to obtain more information for sister clades, as the D and f_4 -ratio statistics are robust even when using whole genome data from

only a single individual per population (Green et al. 2010; Malinsky et al. 2021). To investigate the potential hybrid individual (GLGS1945 from Clade A), we applied the “Ancestry Painting” method, using sites fixed between the two putative parental lineages (Clades A and B).

Ancient introgression was examined using D_{FOIL} statistics (Pease and Hahn 2015). A total of 327 penta-taxon input files consistent with phylogeny were generated. Introgression was evaluated by the proportion of windows across the genome supporting introgression from certain ancient nodes to an outside tip node. The D_{FOIL} analysis procedures are provided in the Supplementary Methods.

The analyses described above detected introgression events between Clades I, J, K and L, as well as a potential distal “ghost” introgression into Clade M. However, within the established phylogeny ((I, J), (K, (L, M))), introgression between non-sister clades—specifically between the IJ clade and K, between IJ and L, or between K and L—can produce signals similar to those of distal ghost introgression into Clade M (Zhang et al. 2024). Considering that heuristic approaches like SNaQ might confound these two scenarios (Pang and Zhang 2024), we further tested for the presence of ghost introgression using BPP’s MSci (multispecies coalescent with introgression) model (Flouri et al. 2018, 2019). We extracted 1,000 non-adjacent loci from the alignments used in the ASTRAL analysis and trimmed them to 800 bp to meet the data requirements for the BPP-MSci analysis. Based on the introgression patterns suggested by D statistic results, we constructed multiple models to test for potential ghost introgression that were indicated by SNaQ analysis. Specifically, we analyzed the phylogeny (H, (IJ, (K, (L, M)))) with the sister clades I and J combined to one

clade, a conservative approach taken to minimize unnecessary model complexity and computational demands. The tested models included: (1) a model with no introgression; (2) a model with only distal ghost introgression ($\text{ghost} \rightarrow \text{M}$); (3) a model for the introgression events between “non-sister clades”, where we tested unidirectional introgression events ($\text{IJ} \rightarrow \text{K}$, $\text{K} \rightarrow \text{IJ}$, $\text{IJ} \rightarrow \text{L}$, $\text{L} \rightarrow \text{IJ}$) as well as a bidirectional introgression, $\text{K} \leftrightarrow \text{L}$; and (4) a model that builds on model (3) by incorporating “ghost $\rightarrow \text{M}$ ” to assess whether the inclusion of ghost introgression significantly improves the model fit. For the simplest model with no introgression, we ran 30,000 MCMC iterations as burn-in and then took 40,000 samples, sampling every 5 iterations. For models that consider introgression events, the MCMC chain length was increased according to the model complexity to ensure adequate sampling. Each run was checked using Tracer v1.7.2 to confirm that all ESS values were above 200. We used marginal likelihood values for model comparison, which were generated by 16 control files with different beta values.

Species Differentiation

We computed pairwise fixation index (F_{ST}) statistics and absolute pairwise genetic distance (D_{XY}) values in sliding windows to assess genetic differentiation among the clades using the popgenWindows.py script from the genomics general package (https://github.com/simonhmartin/genomics_general). In addition, we calculated the proportion of heterozygous sites (H_{ct}) for each individual to investigate genetic diversity.

Isolation-by-distance (IBD) Analysis

To test the potential “ring-like differentiation,” an isolation-by-distance (IBD) pattern was examined for clades around the Sichuan Basin (SB) using R package *vegan* (Dixon 2003). Mantel tests with 1,000 permutations were used to detect significant correlations between genetic distance ($F_{ST}/(1 - F_{ST})$) and geographic distance. “Straight distances” (Euclidean distance) and “ring distances” were calculated using an online coordinate distance calculator (<http://boulter.com/gps/distance/>) (Supplementary Table S3). The “ring distances” assuming dispersal routes following the ring distribution, instead of straight lines across the uninhabitable basin area, were measured with four reference points following Wang et al. (2021) (A, B, C and D, listed in Supplementary Table S1). Based on gene flow tests indicating that a genetic “gap” occurred at the southwest corner of the SB, we assumed that no genetic exchange among different clades occurred across this “gap” when “ring distances” were calculated. Furthermore, *U. andersoni* (Clade D), *U. aequodonenia* (Clade E) and one clade of *U. gracilis* (Clade F) diverged far before differentiation around the SB, and the introgression signal implied that they may not have experienced ring differentiation. Thus, IBD analyses were conducted for clades around the SB, including and excluding these three clades, respectively.

Demographic History Inference

We used the Pairwise Sequentially Markovian Coalescence model (PSMC) (Li and Durbin 2011; Liu and Hansen 2017) to estimate the demographic history of populations. The hybrid individual in Clade A was excluded because its extremely high heterozygosity value

led to outlier results. The parameters were set as follows: $-N25 -t15 -r5; -p '4+25*2+4+6'$.

The mutation rate per generation per site was estimated as $\mu = D * g / 2T$, where D is the observed frequency of pairwise differences between two species, T is the estimated divergence time and g is the estimated generation time for the two species. We used the values between *U. investigator* and other species: D (D_{XY}) = 0.023, T = 5.54 Ma and g was set to 0.5 years. These values yielded an estimated mutation rate of 1.0×10^{-9} mutations per generation per site for *Uropsilus*. The PSMC analysis was implemented with 100 bootstrap replicates to assess the variation in the inferred effective population size (N_e) trajectories.

RESULTS

Phylogenomic Inference and Discordance

The ML tree constructed from the CDS-4dtv dataset with an outgroup (*Talpa occidentalis*) identified *U. investigator* (Clades A and B) as the earliest diverging lineage within genus *Uropsilus* (Supplementary Fig. S2). Phylogenetic analyses employing ML, NJ, and coalescent methods rooted with *U. investigator* revealed identical topologies (Fig. 1b; Supplementary Fig. S3-S5). All nodes exhibited 100% bootstrap support in the ML tree. We identified 16 lineages corresponding to 11 known species and five putative species (Clades A-P) in all trees, including the mitochondrial-based tree (Fig. 1b). *U. gracilis* represents a polyphyletic species group comprising Clades F, I, J and K. We divided the clades into six primary groups (Groups 1-6) based on phylogenetic clustering.

Incongruent topologies were observed between phylogenies inferred from nuclear genomes and mitochondrial genes. The tree constructed using nuclear genomic SNPs placed

Group 4 (Clade F: *U. gracilis*) as the sister branch of Group (5+6), and Clade G (*U. dabieshanensis*) clustered within Group 5. Conversely, in the mitogenomic tree, Group 4 formed the sister branch of Group (2+3), and Clade G formed the sister branch of Group (5+6). Additionally, the relationships among Clade K (*U. gracilis*), Clade L (*U. fansipanensis*), and Clade M (*U. atronates*) show low bootstrap support values for the mitochondrial tree (Fig. 1b).

Among the three types of concordance factors (ψ_1 in CVs), sCFs consistently produced lower values—even for branches where the qCF and gCF approached 100% (Fig. 2). This pattern may be explained by homoplasy at individual sites (Lanfear and Hahn 2024). Notably, qCVs may demonstrate greater robustness to gene tree error than gCVs (Lanfear and Hahn 2024): while gCVs depend on individual gene tree topologies and are therefore sensitive to poorly resolved trees, qCVs utilize quartet information that mitigates uncertainty and enhances stability. The three methods also differ in their treatment of ψ_4 . In both sCVs and qCVs, ψ_4 is fixed at 0 by definition, as only three possible quartet topologies exist. In contrast, gCVs allow ψ_4 to represent the cumulative proportion of gene trees supporting topologies other than the primary (ψ_1) or secondary alternatives (ψ_2 and ψ_3).

Our results show that most ψ_4 values in gCVs remain low—particularly being substantially lower than the proportion of the primary topology (ψ_1)—indicating that non-monophyly occurs in only a small subset of gene trees. The branch with the highest discordance is Clade A, where ψ_2 is nearly equal to or exceeded ψ_1 . Our subsequent introgression and H_{et} analysis indicated that this discordance stems from a hybrid individual in Clade A, resulting from admixture between Clade A and Clade B. Widespread discordance

is most evident in Group 5, indicating a high level of ILS among its constituent clades. Moreover, the imbalance between ψ_2 and ψ_3 , coupled with a relatively high ψ_4 , suggests potential introgression or hybridization among these clades in Group 5 (Fig. 2b). Although notable discordance was also observed in Group 6, closer ψ_2 and ψ_3 implies that ILS, rather than gene flow, is the more likely cause of phylogenetic incongruence in this group.

Divergence Time Estimation

Although the SNAPP and MCMCTree analyses exhibited subtle discrepancies, both consistently supported a late Miocene divergence of *Uropsilus* (5.54 Ma in SNAPP vs. 6.65 Ma in MCMCTree) (Fig. 1c; Supplementary Fig. S6). Both methods support that the divergences among clades are concentrated in the Late Pliocene to the Early Pleistocene (~2.70~1.12 Ma). Considering that SNAPP utilizes a coalescent-based method, its inferences may more closely reflect the true evolutionary history (Fig. 1c).

Phylogenetic Networks, Introgression and Species Differentiation

The phylogenetic network inferred by SNaQ showed a significant decrease in -loglik scores as h_{max} increased from 0 to 1 (Supplementary Fig. S7a). When h_{max} increased further, the scores fluctuated within a relatively narrow range. Although the lowest -loglik score was observed at $h_{max} = 5$, the γ values of the minor hybrid edges, aside from the primary hybridization event inferred at $h_{max} = 1$, were all extremely small (<0.07) (Supplementary Fig. S7b), suggesting they might represent spurious signals or biologically insignificant events.

Therefore, we adopted the results from $h_{max} = 1$. In this hybridization event, the major edge

supports the topology of the species tree ($\gamma=0.68$), while the minor edge suggests that Clade M is sister to the common ancestor of Clade I-L ($\gamma=0.32$) (Fig. 3a).

In the introgression analyses, the highest introgression levels were observed between Clade K (*U. gracilis*) and Clade L (*U. fansipanensis*) ($D = 0.28$; f_4 ratio = 0.06; $f(C) = 0.06$), and between *U. gracilis*'s three clades, Clades I, J and K ($D = 0.20$ - 0.21 ; f_4 ratio = 0.05; $f(C) = 0.05$) (Fig. 3b; Supplementary Fig. S8). These results are consistent with the gene flow implied by discordance factors (Fig. 2b). We also detected significant introgression between Clades I and J, with $fb(C)$ reaching 0.04, and between the two *U. soricipes* clades (Clades O and P) with $fb(C)$ reaching 0.03 (Fig. 2b). Negligible introgression levels were detected among the remaining clades, such as between Clades G (*U. dabieshanensis*) and N-P (*U. gracilis* and *U. soricipes*), and between Clades N (*U. funiushanensis*) and O (*U. soricipes*). The intensity of recent gene flow does not appear to account for the phylogenetic incongruence of Clades F and G. Although gene flow was detected among multiple Group 5 clades, no evidence of gene flow was observed involving the most recently diverged Clade M (*U. atronates*). The D test results based on the quartets of all clades or subclades are largely consistent with the above analysis using *U. investigator* as the outgroup for the main introgression signals (Supplementary Fig. S9). However, considering the potential ancient introgression between *U. investigator* and the ingroup, we avoided interpreting overly weak introgression signals. One exceptionally high introgression signal resulted from a hybrid individual (sample ID: GLGS1945) between Clade A and B in the sympatric area of these two clades in the mid Gaoligong Mountain, as clearly illustrated by ancestry painting and an additional f-branch test for individuals in Clade A and B (Supplementary Fig. S10, S11).

Of the 327 penta-taxon phylogenies eligible for D_{FOIL} analysis, a total of 34 penta-taxon phylogenies exhibited more than 10% windows supporting ancient introgression (Supplementary Table S5). These can be divided into three scenarios (Fig. 3c; Supplementary Table S5): (1) Introgression between Clade N and the ancestors of Clade I-M in Group 5, supported by 11.97-13.18% of the windows; (2) Introgression between Clade G and the ancestors of Clades O, P1, and P2 in Group 6, supported by 10.08-13.15% of the windows; and (3) Introgression between Clade K and the ancestor of Clade (I+J), supported by 11.26% of the windows. Additionally, introgression between Clade F and the ancestor of Clades (D+E) (*U. andersoni* and *U. aequedonenia*) was also detected in 8.17% of the windows. The results of the D_{FOIL} analysis indicated that some weak but statistically significant signals of introgression detected in D , f_4 , and f-branch tests may be attributed to ancestral introgression. In addition, similar to the results of Dsuite analyses, no introgression was detected between Clade M and any clades within Group 5 (Supplementary Table S5).

The results from BPP's MSci model analysis revealed that adding introgression events—whether introgression from a ghost lineage or introgression between “non-sister clades”—was significantly superior to the no-introgression model (Supplementary Fig. S12). Among the single introgression events, the best-fitting model was the IJ→L model. Moreover, for all introgression scenarios between “non-sister clades”, incorporating ghost introgression further improved the model fit (Supplementary Fig. S12). These BPP results support the presence of both ghost introgression and non-sister introgression.

The absolute pairwise genetic distance, D_{XY} , between clades ranges from 0.35% to 2.40% (Supplementary Fig. S13). The lowest D_{XY} between recognized species is 0.39%—between *U.*

fansipanensis (Clade L) and *U. atronates* (Clade M). With the exception of Clades I and J of *U. gracilis* ($D_{XY}=0.35\%$), the D_{XY} between all other clades exceed 0.39%, indicating a high level of differentiation among clades/cryptic species. The pairwise fixation index (F_{ST}) ranges from 0.17-0.91, with the lowest value occurring in Clades O and P of *U. soricipes* (Supplementary Fig. S13). There is a large discrepancy in heterozygous site proportion (H_{et}) among different clades (Supplementary Fig. S14). An exceptionally high H_{et} value was detected in the hybrid individual of Clade A (*U. investigator*).

Isolation-by-distance (IBD) patterns

Isolation-by-distance (IBD) analyses of clades around the SB (Clades D, E, F, I, J, N, O, and P) showed that genetic distance and Euclidean distances (straight distance) were not significantly correlated (Mantel test: $r = -0.32$, $p = 0.992$) (Supplementary Fig. S15a). However, the adjusted ring distance was significantly correlated with genetic distance (Mantel test: $r = 0.30$, $p = 0.004$) (Supplementary Fig. S15b). The Mantel test excluding Clades D (*U. andersoni*), E (*U. aequodonenia*) and F (*U. gracilis*) revealed a significant and stronger correlation between ring distance and genetic distance ($r = 0.85$, $p = 0.001$), whereas Euclidean distance and genetic distance were uncorrelated ($r = -0.02$, $p = 0.51$) (Supplementary Fig. S15c-d).

Demographic History Analysis

The PSMC analysis revealed different demographic histories for each clade of Asian shrew moles (Fig. 4; Supplementary Fig. S16). Different geographically isolated populations

of Clade C (*U. nivatus*) and Clade P (*U. soricipes*) also exhibit distinct demographic dynamics (Fig. 4c, g). However, in Group 1, 2, 3, 4 and 6, population size generally decreased during the Naynayxungla glaciation (NG, 500 - 720 Ka (Sun and An 2005; Chen et al. 2019)). In contrast, Group 5 populations did not experience contraction during the NG (Fig. 4f); instead, they exhibited a diffusion-like profile of the effective population size curve after species divergence. Most clades exhibited population expansion during the last interglacial period (LIG, 129 Ka-116 Ka).

DISCUSSION

Implications of Genomic Phylogenetic Heterogeneity

Mito-nuclear phylogenetic incongruences.—

Mito-nuclear incongruences were found for Clades F (*U. gracilis*) and G (*U. dabieshanensis*), but no significant phylogenetic heterogeneity was observed across the nuclear genome (Fig. 1b; Fig. 2). These two clades diverged from all other clades around 4.26 and 2.49 Ma, respectively, and have since undergone a prolonged period of independent evolution (Fig. 1c); few significant ongoing introgressions have been detected. However, both clades exhibited some evidence of ancient introgression (Fig. 3c). Ancient introgression occurred between Clade F and its parapatric clades, including Clades D (*U. andersoni*) and E (*U. aequodonenia*), possibly due to historical contact. In contrast, Clade G experienced ancient introgression between more distant clades (Clade P: *U. soricipes*). It should be noted that the D_{FOIL} test was performed on 327 five-taxon subsets from the entire tree (Pease and Hahn 2015). If these subsets share branches on the phylogeny, a single introgression event

may lead to a correlated introgression signal (Malinsky et al. 2021). This may explain the introgression detected between Clade G and multiple ancestral nodes of Clades N, O, P1 and P2, even if only one of these signals corresponds to a true introgression event. A similar pattern was also observed between Clade N and Group 5 lineages. Nevertheless, the introgression between Clade G and Clade P could also indicate historical long-distance dispersal, although this requires more extensive sampling and detailed historical biogeographic analyses for confirmation.

Previous phylogenetic studies of Asian shrew moles based on a limited number of gene fragments often incorporated one or two mitochondrial genes (Wan et al. 2018; Hu et al. 2021; Ren et al. 2023). Due to their faster evolutionary rates, mitochondrial genes may better support the monophyly of species, which is important for species classification. However, when the goal is to construct a robust species tree, ILS as well as the strong phylogenetic signal from mitochondrial genes in datasets with only a few nuclear gene fragments can lead to misleading or spurious topologies (Ballard and Whitlock 2004; Zou and Ge 2008; Toews and Brelsford 2012; Andersen et al. 2021). Nonetheless, mito-nuclear incongruences may still carry valuable evolutionary insights, potentially serving as a starting point for addressing deeper issues such as complex biogeographic histories (Duan et al. 2023; Musher et al. 2023; Rivas-González et al. 2023).

ILS, introgression and ghost introgression.—

In contrast to the pattern of mito-nuclear conflict without genome-wide ILS or introgression described above, we found high levels of both gene flow and ILS across the

nuclear genome within Clades I-L, as well as evidence for ancestral introgression between Clades K and the ancestor of Clades (I+J) (Fig. 2; Fig. 3b-c). These lineages have undergone rapid radiative diversification. Our phylogenetic network analysis also identified a significant hybridization event involving Clade M, with the donor lineage branching from the most recent common ancestor (MRCA) of Clade I-M. Interestingly, no gene flow was detected between the “recipient” Clade M and its sister clades I, J, K or L. We also observed that, compared to its sister clades, Clade M exhibited abnormally long branches in the coalescent-based tree and a high degree of genomic “concordance” (Supplementary Fig. 3). These observations collectively suggest that the detected hybridization events may be the result of “ghost” introgression (Fig. 3a), as gene flow from a distal, unsampled “donor” lineage can lead to unusual divergence between the “recipient” clade (Clade M) and the ingroup sister taxon (Clades I, J, K and L) (Pease and Hahn 2015).

However, as described in the related Methods section, the significant introgression detected among these “non-sister clades” (Clade I, J, K and L) could produce *D*-statistic patterns similar to those expected from distal ghost introgression (Zhang et al. 2024), potentially misleading SNaQ’s inference as well. The BPP methods, which takes into account both gene-tree topologies and branch lengths, has shown an improved ability to distinguish between these two scenarios (Pang and Zhang 2024). Our BPP analysis strongly supports the hypothesis that both ghost introgression and introgression between “non-sister clades” have occurred. Importantly, the gene flow patterns among these lineages in Group 5 might be far more complex than our model assumptions. The BPP-MSci model assumes that gene introgression occurs as discrete events rather than continuously (Flouri et al. 2019;

Thawornwattana et al. 2022). To test for ghost introgression while minimizing computational costs and reducing potential biases in parameter estimation from overparameterized models, we developed simplified model frameworks in this study.

Taxonomic implications.—

Although introgression was detected between some clades (putative species) of Asian shrew moles, all of them exhibited a high degree of differentiation (Supplementary Fig. S13). The D_{XY} values between all clades, except for Clade I and Clade J (*U. gracilis*), are higher than between the two accepted species, *U. fansipanensis* and *U. atronate*. Most of the putative species show a distinct isolated distribution pattern among different “sky islands” (Supplementary Fig. S1). However, the phylogenetic incongruence resulting from their history of rapid radiation may have previously hindered the determination of their species status. The current taxonomic classification is problematic; not only has it failed to include these cryptic species, but it also considered the polyphyly including clades F, I, J and K as *U. gracilis*. Integrating robust species trees (Supplementary Fig. S4), D_{XY} values (0.35-2.40%), and divergence times (1.09-5.54 Ma), our study provides strong genetic evidence supporting the recognition of all five allopatrically distributed putative species as distinct species. However, another factor complicating taxonomic research in this genus is the extreme morphological conservatism among species (Hoffmann 1984; Wan et al. 2018). In future studies, it will be essential to clarify the taxonomy of this group using integrative approaches (DAYRAT 2005; Schlick-Steiner et al. 2010).

Species Diversification in the Context of Sky Island Topography and Climate Change

Compared to other mountain ranges in the Mountains of Southwest China and its adjacent mountains (MSCAM), the Hengduan Mountains (HDMs) region exhibits denser sky islands and steeper elevational gradients, resulting in some of the most pronounced vertical environmental heterogeneity and peak isolation in the region (He and Jiang 2014; Ding et al. 2020). The uplift of the HDMs since the late Miocene (Xing and Ree 2017; Ding et al. 2020) may have driven the earliest divergence of Asian shrew moles around 5.54 Ma (Fig. 1c). Notably, introgression events were generally rare within the HDMs region, with virtually no detectable gene flow even among sympatrically distributed species (Fig. 3d). An exception was observed in a small area of sympatry between Clade A and Clade B in the Gaoligong Mountains, where we identified a hybrid individual (illustrated by ancestry painting in Supplementary Fig. S10). However, the two clades have clearly undergone significant divergence (Supplementary Fig. S11, S13 and S14). When excluding the hybrid individual, the genetic distance between Clade A and B reached 0.62%, which exceeds that observed between any other pair of sister species. Before new *Uropsilus* species were discovered in the East China region (Hu et al. 2021; Ren et al. 2023), most known species were distributed within the HDMs region, including the mountainous areas on the eastern side of the Sichuan Basin. Therefore, vicariance events have long been considered the primary speciation driver in this group (Wan et al. 2018).

By contrast, our study detected widespread gene flow among species in regions east of the HDMs. This may be due to these regions having relatively lower elevations, resulting in weaker barriers to dispersal (Fig. 1a). Rapid dispersal and radiative speciation occurred in

Group 5, giving rise to nearly all the lineages south of the Yangtze River in China, including *U. huanggangensis*, *U. gracilis*, *U. fansipanensis* and *U. atronates*. Notably, unlike the other groups, Group 5 did not experience a population decrease during the Naynayxungla glaciation (Fig. 4). The increased intensity of summer monsoon in East Asia, driven by global climate change and rapid mountain uplift, led to a significant rise in precipitation in southern China during this period (An et al. 2001; Xie and Liu 2020; He et al. 2024), which likely lessened the impact of severe glacial environments on the southern populations (Cheng et al. 2016). Surveys in these areas remain insufficient, and the true diversity of Asian shrew moles may have been overlooked, especially considering that we also detected ghost introgression.

In addition to the strong introgression signals observed in Group 5, we also detected relatively weaker introgression or ancient gene flow among the clades exhibiting a ring-like distribution around the Sichuan Basin (Fig. 3). The isolation-by-distance analysis revealed a strong correlation between genetic distances and ring distances (Supplementary Fig. S15), suggesting a ring-like diversification pattern around the Sichuan Basin, similar to what has been observed in various biological groups including odorous frog (Qiao et al. 2018), golden-thread nanmu tree (Xiao et al. 2020), and field mouse (Wang et al. 2021). However, compared to the intraspecific ring-like distribution patterns of these other species groups, the ring-like distribution in Asian shrew moles is characterized by relatively stronger isolation among populations. This pattern resembles the “stepping-stone” model observed in long-tail moles (Hannah et al. 2014; He et al. 2019), in which populations around the Sichuan Basin form isolated “islands”, yet gene flow may still occur between adjacent ones. Notably, in both long-tail moles and Asian shrew moles, gene flow—possibly ancient—was detected across

the Yangtze River, despite populations north and south of the river having diverged around 3.30 Ma in the latter (Fig. 1c; Fig. 2d). These gene flow patterns suggest that the mountainous regions surrounding the Sichuan Basin may have served as corridors for interspecies genetic exchange in Asian shrew moles, though such exchanges were more likely to occur in the past rather than at present. Additionally, while PSMC revealed distinct demographic histories among different populations, the demographic trajectories of adjacent sister taxa around the Sichuan Basin remained similar for a period following their divergence (Fig. 4). For instance, between Clades N, O, and P, as well as between Clades I and J, population dynamics only began to diverge 200,000-500,000 years ago. This may also imply continued gene flow between these populations even after their initial divergence indicated by SNAPP (Fig. 1c). Given the limited sampling, we cannot yet draw a definitive conclusion regarding the formation of a ring species (Huxley 1943; Irwin et al. 2001). Nonetheless, our findings underscore the need for expanded surveys to the east of the HDMs and a deeper investigation of the region's biogeographic history.

SUPPLEMENTARY MATERIAL

The supplementary file including supplementary methods, figures and tables is available at SYSBIO online.

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AUTHOR CONTRIBUTIONS

X.L.J. conceptualized and designed the study, provided overall guidance and supervision, and critically revised the manuscript. Y.X.L. conducted data analysis, interpreted the results, and wrote the original draft. Z.Z.C. assisted in data interpretation and manuscript review. Y.X.L., Z.Z.C., T.Z., F.C., and K.O.O. contributed to the revision and editing of the manuscript during the peer review process. Z.Z.C., Q.L., W.Y.S. and X.Y.L. collected the field samples. Y.X.L., S.W.H. and H.J.W. conducted the laboratory work. All authors have read and approved the final manuscript.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

DATA AVAILABILITY

The whole genome resequencing data reported in this study have been deposited in the NCBI Sequence Read Archive (SRA) under Bioproject accession number PRJNA1187160. The mitogenome sequences assembled in this study have been deposited in GenBank under accession numbers PV938352-PV938355 and PV939677-PV939711. All other data necessary to reproduce the analyses reported in this study can be accessed through the Dryad link: <https://doi.org/10.5061/dryad.t1g1jwte9>.

REFERENCES

- An Z., John E. K., Warren L. P., Stephen C. P. 2001. Evolution of Asian monsoons and phased uplift of the Himalaya–Tibetan plateau since Late Miocene times. *Nature*, 411:62-66.
- Ballard J.W.O., Whitlock M.C. 2004. The incomplete natural history of mitochondria. *Mol. Ecol.*, 13.
- Bryant D., Bouckaert R., Felsenstein J., Rosenberg N.A., RoyChoudhury A. 2012. Inferring species trees directly from biallelic genetic markers: bypassing gene trees in a full coalescent analysis. *Mol. Biol. Evol.*, 29:1917-1932.
- Bui H.T., Okabe S., Le L.T.H., Ngan T.N., Motokawa M. 2023. A new shrew mole species of the genus *Uropsilus* (Eulipotyphla: Talpidae) from northwestern Vietnam. *Zootaxa*, 5339:059–078.
- Chang C.C., Chow C.C., Tellier L.C., Vattikuti S., Purcell S.M., Lee J.J. 2015. Second-generation PLINK: rising to the challenge of larger and richer datasets. *GigaScience*, 4.
- Chen J., Hao Z., Guang X., Zhao C., Wang P., Xue L., Zhu Q., Yang L., Sheng Y., Zhou Y., Xu H., Xie H., Long X., Zhang J., Wang Z., Shi M., Lu Y., Liu S., Guan L., Zhu Q., Yang L., Ge S., Cheng T., Laux T., Gao Q., Peng Y., Liu N., Yang S., Shi J. 2019. *Liriodendron* genome sheds light on angiosperm phylogeny and species–pair differentiation. *Nature Plants*, 5:18-25.
- Chen S., Zhou Y., Chen Y., Gu J. 2018. fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34:i884-i890.
- Cheng H., Edwards R.L., Sinha A., Spötl C., Yi L., Chen S., Kelly M., Kathayat G., Wang X., Li X., Kong

- X., Wang Y., Ning Y., Zhang H. 2016. The Asian monsoon over the past 640,000 years and ice age terminations. *Nature*, 534:640-646.
- Danecek P., Auton A., Abecasis G., Albers C.A., Banks E., DePristo M.A., Handsaker R.E., Lunter G., Marth G.T., Sherry S.T., McVean G., Durbin R. 2011. The variant call format and VCFtools. *Bioinformatics*, 27:2156-2158.
- DAYRAT B. 2005. Towards integrative taxonomy. *Biol. J. Linn. Soc.*, 85:407-417.
- Degnan J.H., Rosenberg N.A. 2009. Gene tree discordance, phylogenetic inference and the multispecies coalescent. *Trends Ecol. Evol.*, 24:332-340.
- Ding W.N., Ree R.H., Spicer R.A., Xing Y.W. 2020. Ancient orogenic and monsoon-driven assembly of the world's richest temperate alpine flora. *Science*, 369:578-581.
- Dixon P. 2003. VEGAN, a package of R functions for community ecology. *J. Veg. Sci.*, 14:927-930.
- Dong F., Li S.H., Chiu C.C., Dong L., Yao C.T., Yang X.J. 2020. Strict allopatric speciation of sky island *Pyrrhula erythaca* species complex. *Mol. Phylogen. Evol.*, 153:106941.
- Douady C.J., Douzery E.J.P. 2003. Molecular estimation of eulipotyphlan divergence times and the evolution of "Insectivora". *Mol. Phylogen. Evol.*, 28:285-296.
- Duan L., Fu L., Chen H.F. 2023. Phylogenomic cytonuclear discordance and evolutionary histories of plants and animals. *Sci. China Life Sci.*, 66:2946-2948.
- Emms D.M., Kelly S. 2019. OrthoFinder: phylogenetic orthology inference for comparative genomics. *Genome Biol.*, 20:238.
- Felsenstein J. 2009. PHYLIP (Phylogeny Inference Package) version 3.7a. Distributed by the author. Department of Genome Sciences, University of Washington, Seattle.
- Flouri T., Jiao X., Rannala B., Yang Z. 2018. Species Tree Inference with BPP Using Genomic Sequences and the Multispecies Coalescent. *Mol. Biol. Evol.*, 35:2585-2593.
- Flouri T., Jiao X., Rannala B., Yang Z. 2019. A Bayesian Implementation of the Multispecies Coalescent Model with Introgression for Phylogenomic Analysis. *Mol. Biol. Evol.*, 37:1211-1223.
- García-Alix A., Furió M., Minwer-Barakat R., Martín Suárez E., Freudenthal M. 2011. Environmental control on the biogeographical distribution of *Desmanella* (Soricomorpha, Mammalia) in the Miocene of the Iberian Peninsula. *Palaeontology*, 54:753-762.
- Green R.E., Krause J., Briggs A.W., Maricic T., Stenzel U., Kircher M., Patterson N., Li H., Zhai W., Fritz M.H.Y., Hansen N.F., Durand E.Y., Malaspinas A.-S., Jensen J.D., Marques-Bonet T., Alkan C., Prüfer

- K., Meyer M., Burbano H.A., Good J.M., Schultz R., Aximu-Petri A., Butthof A., Höber B., Höffner B., Siegemund M., Weihmann A., Nusbaum C., Lander E.S., Russ C., Novod N., Affourtit J., Egholm M., Verna C., Rudan P., Brajkovic D., Kucan Ž., Gušić I., Doronichev V.B., Golovanova L.V., Lalueza-Fox C., de la Rasilla M., Fortea J., Rosas A., Schmitz R.W., Johnson P.L.F., Eichler E.E., Falush D., Birney E., Mullikin J.C., Slatkin M., Nielsen R., Kelso J., Lachmann M., Reich D., Pääbo S. 2010. A Draft Sequence of the Neandertal Genome. *Science*, 328:710-722.
- He K., Eastman T.G., Czolacz H., Li S., Shinohara A., Kawada S.-i., Springer M.S., Berenbrink M., Campbell K.L. 2021. Myoglobin primary structure reveals multiple convergent transitions to semi-aquatic life in the world's smallest mammalian divers. *eLife*, 10:e66797.
- He K., Gutiérrez E.E., Heming N.M., Koepfli K.P., Wan T., He S., Jin W., Liu S.Y., Jiang X.L. 2019. Cryptic phylogeographic history sheds light on the generation of species diversity in sky-island mountains. *J. Biogeogr.*, 46:2232-2247.
- He K., Jiang X.L. 2014. Sky islands of southwest China. I: an overview of phylogeographic patterns. *Chin. Sci. Bull.*, 59:585-597.
- He K., Shinohara A., Helgen K.M., Springer M.S., Jiang X.L., Campbell K.L. 2016. Talpid Mole Phylogeny Unites Shrew Moles and Illuminates Overlooked Cryptic Species Diversity. *Mol. Biol. Evol.*, 34:78-87.
- He L., Zhou T., Guo Z., Zuo M., Ren Z., Chen X., Wu B., Zou L., Zhang L., Man W., Jiang J. 2024. Northward Extension of East Asian Summer Monsoon Since the Miocene Set by the Uplift of Tibetan Plateau. *Geophys. Res. Lett.*, 51:e2023GL107262.
- Hannah L., Flint L., Syphard A.D., Moritz M.A., Buckley L.B., McCullough I.M. 2014. Fine-grain modeling of species' response to climate change: holdouts, stepping-stones, and microrefugia. *Trends Ecol. Evol.*, 29:390-397.
- Helfrich P., Rieb E., Abrami G., Lücking A., Mehler A. 2018. TreeAnnotator: Versatile Visual Annotation of Hierarchical Text Relations. *International Conference on Language Resources and Evaluation*.
- Hoffmann R.S. 1984. A review of the shrew-moles (genus *Uropsilus*) of China and Burma. *J. Mamm. Soc. Japan*, 10:69-80.
- Hu T.L., Xu Z., Zhang H., Liu Y.X., Liao R., Yang G.D., Sun R.L., Shi J., Ban Q., Li C.L., Liu S.Y., Zhang B.W. 2021. Description of a new species of the genus *Uropsilus* (Eulipotyphla: Talpidae: Uropsilinae) from the Dabie Mountains, Anhui, Eastern China. *Zool. Res.*, 42:294-299.

- Huxley J. 1943. Systematics and the origin of Species : from the Viewpoint of a Zoologist. *Nature*, 151:347-348.
- Irwin D.E., Irwin J.H., Price T.D. 2001. Ring species as bridges between microevolution and speciation. *Genetica*, 112:223-243.
- Jiang H.J., Yang Y.H., Zhou Y.Y., Tian J.D., Lu J.Q. 2025. Molecular Phylogeny of *Uropsilus* (Talpidae, Eulipotyphla, Mammalia) With a New Species Described From Henan Province, China. *Ecol. Evol.*, 15:e70928.
- Jin J.J., Yu W.B., Yang J.B., Song Y., dePamphilis C.W., Yi T.-S., Li D.-Z. 2020. GetOrganelle: a fast and versatile toolkit for accurate de novo assembly of organelle genomes. *Genome Biol.*, 21:241.
- Kalyaanamoorthy S., Minh B.Q., Wong T.K.F., von Haeseler A., Jermini L.S. 2017. ModelFinder: fast model selection for accurate phylogenetic estimates. *Nat. Methods*, 14:587-589.
- Klietmann J., Metscher B.D., van den Hoek Ostende L.W., Nagel D., Rummel M. 2013. First record of an upper deciduous molar in *Desmanella* (Uropsilinae, Talpidae, Mammalia). *Geobios*, 46:503-510.
- Korneliussen T.S., Albrechtsen A., Nielsen R. 2014. ANGSD: Analysis of Next Generation Sequencing Data. *BMC Bioinf.*, 15:356.
- Lanfear R., Frandsen P.B., Wright A.M., Senfeld T., Calcott B. 2017. PartitionFinder 2: New Methods for Selecting Partitioned Models of Evolution for Molecular and Morphological Phylogenetic Analyses. *Mol. Biol. Evol.*, 34:772-773.
- Lanfear R., Hahn M.W. 2024. The Meaning and Measure of Concordance Factors in Phylogenomics. *Mol. Biol. Evol.*, 41.
- Larkin M.A., Blackshields G., Brown N.P., Chenna R., McGettigan P.A., McWilliam H., Valentin F., Wallace I.M., Wilm A., Lopez R., Thompson J.D., Gibson T.J., Higgins D.G. 2007. Clustal W and Clustal X version 2.0. *Bioinformatics*, 23:2947-2948.
- Leaché A.D., Banbury B.L., Felsenstein J., de Oca A.n.-M., Stamatakis A. 2015. Short Tree, Long Tree, Right Tree, Wrong Tree: New Acquisition Bias Corrections for Inferring SNP Phylogenies. *Syst. Biol.*, 64:1032-1047.
- Lei F.M., Qu Y.H., Song G., Alstrom P., Fjeldsa J. 2015. The potential drivers in forming avian biodiversity hotspots in the East Himalaya Mountains of Southwest China. *Integr. Zool.*, 10:171-181.
- Li H. 2013. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. *arXiv*:

Genomics.

- Li H., Durbin R. 2011. Inference of human population history from individual whole-genome sequences. *Nature*, 475:493.
- Liu Q., Hua Q.Y., Sun R., He S.W., Li M.C., Shi L.Y., Chen P., Ma Y.S., Zhang Q., Jiang X.L., Wang Y., Shi P. 2023. Evolutionary prediction for new echolocators. *bioRxiv:2023.2009.2013.556757*.
- Liu S., Hansen M.M. 2017. PSMC (pairwise sequentially Markovian coalescent) analysis of RAD (restriction site associated DNA) sequencing data. *Mol. Ecol. Resour.*, 17:631-641.
- Malinsky M., Matschiner M., Svardal H. 2021. Dsuite - Fast D-statistics and related admixture evidence from VCF files. *Mol. Ecol. Resour.*, 21:584-595.
- Marchese C. 2015. Biodiversity hotspots: A shortcut for a more complicated concept. *Global Ecol. Conserv.*, 3:297-309.
- McKenna A., Hanna M., Banks E., Sivachenko A., Cibulskis K., Kernytsky A., Garimella K., Altshuler D., Gabriel S., Daly M., DePristo M.A. 2010. The Genome Analysis Toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. *Genome Res.*, 20:1297-1303.
- Meleshko O., Martin M.D., Korneliussen T.S., Schröck C., Lamkowski P., Schmutz J., Healey A., Piatkowski B.T., Shaw A.J., Weston D.J., Flatberg K.I., Szövényi P., Hassel K., Stenøien H.K. 2021. Extensive Genome-Wide Phylogenetic Discordance Is Due to Incomplete Lineage Sorting and Not Ongoing Introgression in a Rapidly Radiated Bryophyte Genus. *Mol. Biol. Evol.*, 38:2750-2766.
- Meng G., Li Y., Yang C., Liu S. 2019. MitoZ: a toolkit for animal mitochondrial genome assembly, annotation and visualization. *Nucleic Acids Res.*, 47:e63-e63.
- Meyer B.S., Matschiner M., Salzburger W. 2016. Disentangling Incomplete Lineage Sorting and Introgression to Refine Species-Tree Estimates for Lake Tanganyika Cichlid Fishes. *Syst. Biol.*, 66:531-550.
- Minh B.Q., Schmidt H.A., Chernomor O., Schrempf D., Woodhams M.D., von Haeseler A., Lanfear R. 2020. IQ-TREE 2: New Models and Efficient Methods for Phylogenetic Inference in the Genomic Era. *Mol. Biol. Evol.*, 37:1530-1534.
- Musher L.J., Del-Rio G., Marcondes R.S., Brumfield R.T., Bravo G.A., Thom G. 2023. Geogenomic Predictors of Genetree Heterogeneity Explain Phylogeographic and Introgression History: A Case Study in an Amazonian Bird (*Thamnophilus aethiops*). *Syst. Biol.*, 73:36-52.
- Myers N., Mittermeier R.A., Mittermeier C.G., da Fonseca G.A., Kent J. 2000. Biodiversity hotspots for

conservation priorities. *Nature*, 403:853-858.

Olave M., Meyer A. 2020. Implementing Large Genomic Single Nucleotide Polymorphism Data Sets in Phylogenetic Network Reconstructions: A Case Study of Particularly Rapid Radiations of Cichlid Fish. *Syst. Biol.*, 69:848-862.

Pang X.X., Zhang D.Y. 2024. Detection of Ghost Introgression Requires Exploiting Topological and Branch Length Information. *Syst. Biol.*, 73:207-222.

Pavón-Vázquez C.J., Rana Q., Farleigh K., Crispo E., Zeng M., Lilliah J., Mulcahy D., Ascanio A., Jezkova T., Leaché A.D., Flouri T., Yang Z., Blair C. 2024. Gene Flow and Isolation in the Arid Nearctic Revealed by Genomic Analyses of Desert Spiny Lizards. *Syst. Biol.*, 73:323-342.

Pease J.B., Hahn M.W. 2015. Detection and Polarization of Introgression in a Five-Taxon Phylogeny. *Syst. Biol.*, 64:651-662.

Perrigo A., Hoorn C., Antonelli A. 2020. Why mountains matter for biodiversity. *J. Biogeogr.*, 47:315-325.

Perteu G., Perteu M. 2020. GFF Utilities: GffRead and GffCompare. *F1000Res*, 9.

Qiao L., Wen G., Qi Y., Lu B., Hu J., Song Z., Fu J. 2018. Evolutionary melting pots and reproductive isolation: A ring-shaped diversification of an odorous frog (*Odorrana margaratea*) around the Sichuan Basin. *Mol. Ecol.*, 27:4888-4900.

Qiu Z., Storch G. 2005. The Fossil Record of the Eurasian Neogene Insectivores (Erinaceomorpha, Soricomorpha, Mammalia), Part I: China. *Scripta Geologica. Special Issue*, 5:37-50.

Rahbek C., Borregaard M.K., Antonelli A., Colwell R.K., Holt B.G., Nogues-Bravo D., Rasmussen C.M.Ø., Richardson K., Rosing M.T., Whittaker R.J., Fjeldså J. 2019a. Building mountain biodiversity: Geological and evolutionary processes. *Science*, 365:1114.

Rahbek C., Borregaard M.K., Colwell R.K., Dalsgaard B., Holt B.G., Morueta-Holme N., Nogues-Bravo D., Whittaker R.J., Fjeldså J. 2019b. Humboldt's enigma: What causes global patterns of mountain biodiversity? *Science*, 365:1108.

Rambaut A., Drummond A.J., Xie D., Baele G., Suchard M.A. 2018. Posterior Summarization in Bayesian Phylogenetics Using Tracer 1.7. *Syst. Biol.*, 67:901-904.

Ren X.Y., Xu Y.F., Li Y.X., Yao H.F., Fang Y., Khanal L., Cheng L., Zeng W., Jiang X.L., Chen Z.Z. 2023. A new species of shrew moles, genus *Uropsilus* Milne-Edwards, 1871 (Mammalia, Eulipotyphla, Talpidae), from the Wuyi Mountains, Jiangxi Province, eastern China. *ZooKeys*, 1186:25-46.

Rivas-González I., Rousselle M., Li F., Zhou L., Dutheil J.Y., Munch K., Shao Y., Wu D., Schierup M.H.,

- Zhang G. 2023. Pervasive incomplete lineage sorting illuminates speciation and selection in primates. *Science*, 380:eabn4409.
- Sanchez-Villagra M.R., Horovitz I., Motokawa M. 2006. A comprehensive morphological analysis of talpid moles (Mammalia) phylogenetic relationships. *Cladistics*, 22:59-88.
- Schlick-Steiner B.C., Steiner F.M., Seifert B., Stauffer C., Christian E., Crozier R.H. 2010. Integrative taxonomy: a multisource approach to exploring biodiversity. *Annu. Rev. Entomol.*, 55:421-438.
- Solís-Lemus C., Ané C. 2016. Inferring Phylogenetic Networks with Maximum Pseudolikelihood under Incomplete Lineage Sorting. *PLoS Genet.*, 12:e1005896.
- Springer M.S., Murphy W.J., Roca A.L. 2018. Appropriate fossil calibrations and tree constraints uphold the Mesozoic divergence of solenodons from other extant mammals. *Mol. Phylogen. Evol.*, 121:158-165.
- Stamatakis A. 2014. RAxML version 8: a tool for phylogenetic analysis and post-analysis of large phylogenies. *Bioinformatics*, 30:1312-1313.
- Sun Y., An Z. 2005. Late Pliocene-Pleistocene changes in mass accumulation rates of eolian deposits on the central Chinese Loess Plateau. *J. Geophys. Res.: Atmos.*, 110.
- Thawornwattana Y., Seixas F.A., Yang Z., Mallet J. 2022. Full-Likelihood Genomic Analysis Clarifies a Complex History of Species Divergence and Introgression: The Example of the erato-sara Group of *Heliconius* Butterflies. *Syst. Biol.*, 71:1159-1177.
- Toews D.P.L., Brelsford A. 2012. The biogeography of mitochondrial and nuclear discordance in animals. *Mol. Ecol.*, 21:3907-3930.
- Tricou T., Tannier E., de Vienne D.M. 2022. Ghost Lineages Highly Influence the Interpretation of Introgression Tests. *Syst. Biol.*, 71:1147-1158.
- Wan T., He K., Jiang X.L. 2013. Multilocus phylogeny and cryptic diversity in Asian shrew-like moles (*Uropsilus*, Talpidae): implications for taxonomy and conservation. *BMC Evol. Biol.*, 13:232-232.
- Wan T., He K., Jin W., Liu S.Y., Chen Z.Z., Zhang B., Murphy R.W., Jiang X.L. 2018. Climate niche conservatism and complex topography illuminate the cryptic diversification of Asian shrew-like moles. *J. Biogeogr.*, 45:2400-2414.
- Wang Y., Feijó A., Cheng J., Xia L., Wen Z., Ge D., Sun J., Lu L., Li S., Yang Q. 2021. Ring distribution patterns—diversification or speciation? Comparative phylogeography of two small mammals in the mountains surrounding the Sichuan Basin. *Mol. Ecol.*, 30:2641-2658.

- Xiao J.H., Ding X., Li L., Ma H., Ci X.Q., van der Merwe M., Conran J.G., Li J. 2020. Miocene diversification of a golden-thread nanmu tree species (*Phoebe zhennan*, Lauraceae) around the Sichuan Basin shaped by the East Asian monsoon. *Ecol. Evol.*, 10:10543-10557.
- Xie X.X., Liu X.D. 2020. Regional differences of East Asian monsoon precipitation in response to orbital forcing and global ice-volume changes since the late Middle Pleistocene. *Quat. Sci.*, 40:1486-1498.
- Xing Y.W., Ree R. 2017. Uplift-driven diversification in the Hengduan Mountains, a temperate biodiversity hotspot. *Proc. Natl. Acad. Sci. USA*, 114.
- Yang Y., Tian K., Hao J., Pei S., Yang Y. 2004. Biodiversity and biodiversity conservation in Yunnan, China. *Biodivers. Conserv.*, 13:813-826.
- Yang Z. 2007. PAML 4: Phylogenetic Analysis by Maximum Likelihood. *Mol. Biol. Evol.*, 24:1586-1591.
- Zhang C., Rabiee M., Sayyari E., Mirarab S. 2018. ASTRAL-III: polynomial time species tree reconstruction from partially resolved gene trees. *BMC Bioinf.*, 19:153.
- Zhang R.Z. 2002. Geological events and mammalian distribution in China. *Chin. J. Zool.*, 48:141-153 (in Chinese).
- Zhang W.P., Ding Y.M., Cao Y., Li P., Yang Y., Pang X.X., Bai W.N., Zhang D.Y. 2024. Uncovering ghost introgression through genomic analysis of a distinct eastern Asian hickory species. *The Plant Journal*, 119:1386-1399.
- Zou X.H., Ge S. 2008. Conflicting gene trees and phylogenomics. *J. Syst. Evol.*:795-807 (in Chinese).

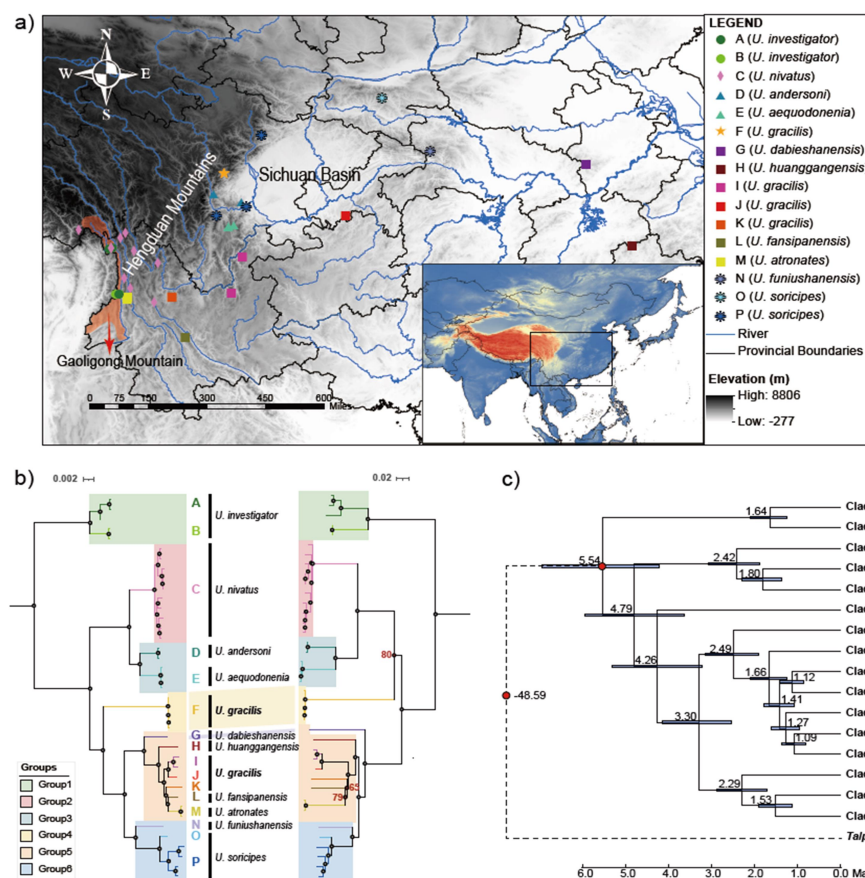


Figure 1 Sampling sites and phylogenetic relationships of 16 clades of Asian shrew moles, genus *Uropsilus* included in this study. a) Sample sites of 39 individuals. Different shapes represent different groups. b) Phylogenetic tree constructed using ML methods based on nuclear-genomic SNPs (left) and mitogenome (right). Interspecific nodes with bootstrap support values below 95 are labelled with corresponding values. Branches in different colors represent different clades. Between the two topologies, vertical black bars represent named species acknowledged in the taxonomic classification, and inconsistency in topology are linked. c) Divergence times estimated by SNAPP, with the node bars representing 95% confidence intervals. Two calibration points are indicated by red dots. Note that the outgroup branch (indicated by dashed lines) has been artificially shortened and its length does not represent the actual divergence time.

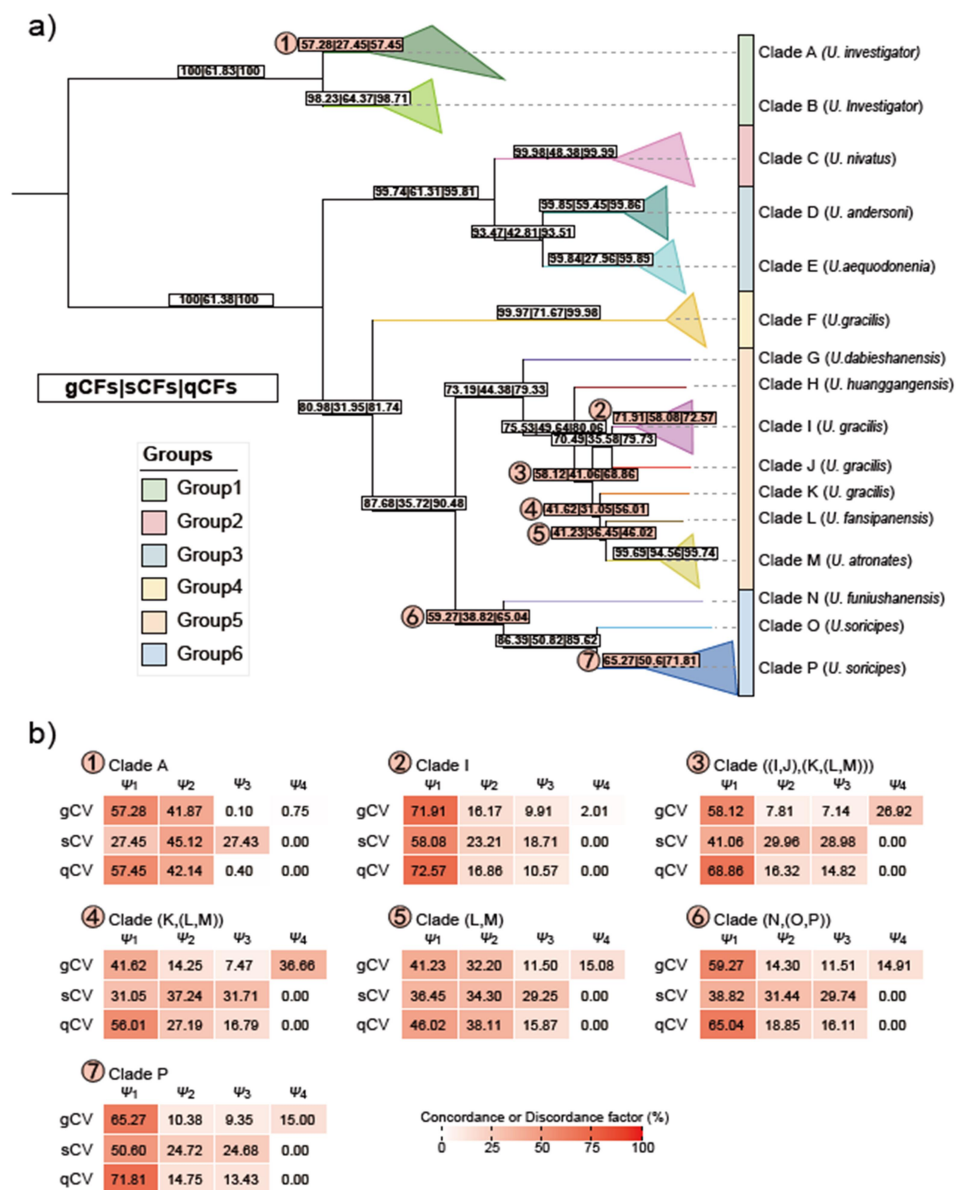


Figure 2 Concordance vector analysis for sixteen clades of Asian shrew moles, genus *Uropsilus*. a) Three concordance factors estimated using ASTRAL and IQ-TREE for branches: gene concordance factors (gCFs), site concordance factors (sCFs), and quartet concordance factors (qCFs). b) The concordance vectors (CVs) of the seven branches with most discordance (numbered as in panel a). ψ_1 : concordance factor; ψ_2 : the largest of the two discordance factors; ψ_3 : the smallest of the two discordance factors; ψ_4 : the sum of all remaining discordance factors not included in ψ_2 or ψ_3 , always zero for sCVs and qCVs.

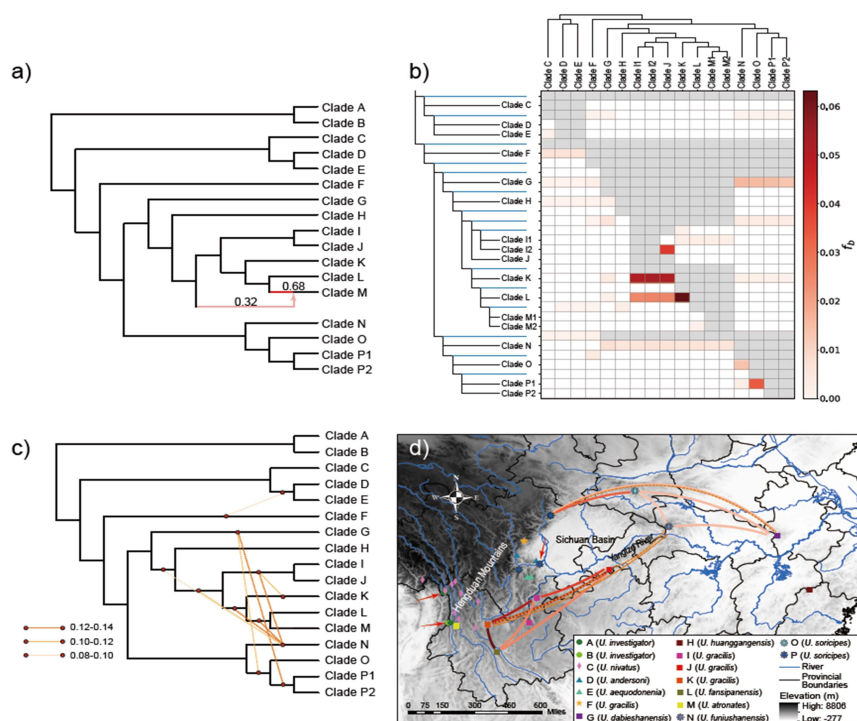


Figure 3 Phylogenetic network and gene flow analysis of Asian shrew moles, genus *Uropsilus*. a) Best phylogenetic relationship inferred from the SNaQ (Species Networks applying Quartets) analysis. The red edge and light-red arrow represent the major and minor edges, respectively, of the inferred hybridization event, with the associated numbers indicating their inheritance probabilities (γ). b) Result of f-branch analysis showing the introgression among different clades. The values in the matrix refer to excess allele shared between the branches identified on the expanded tree on the y axis (relative to its sister branch) and the species identified on the x axis. c) Primary ancient introgression inferred by D_{FOIL} statistics. The color and thickness of the lines indicate the proportion of windows supporting introgression between the branches, as shown on the left. d) Distribution map showing main introgression events among clades. The red arrows indicate the locations where different clades are sympatric. The line styles (dashed/solid) and colors correspond to the f-branch results in (b) and the D_{FOIL} detection results in (c).

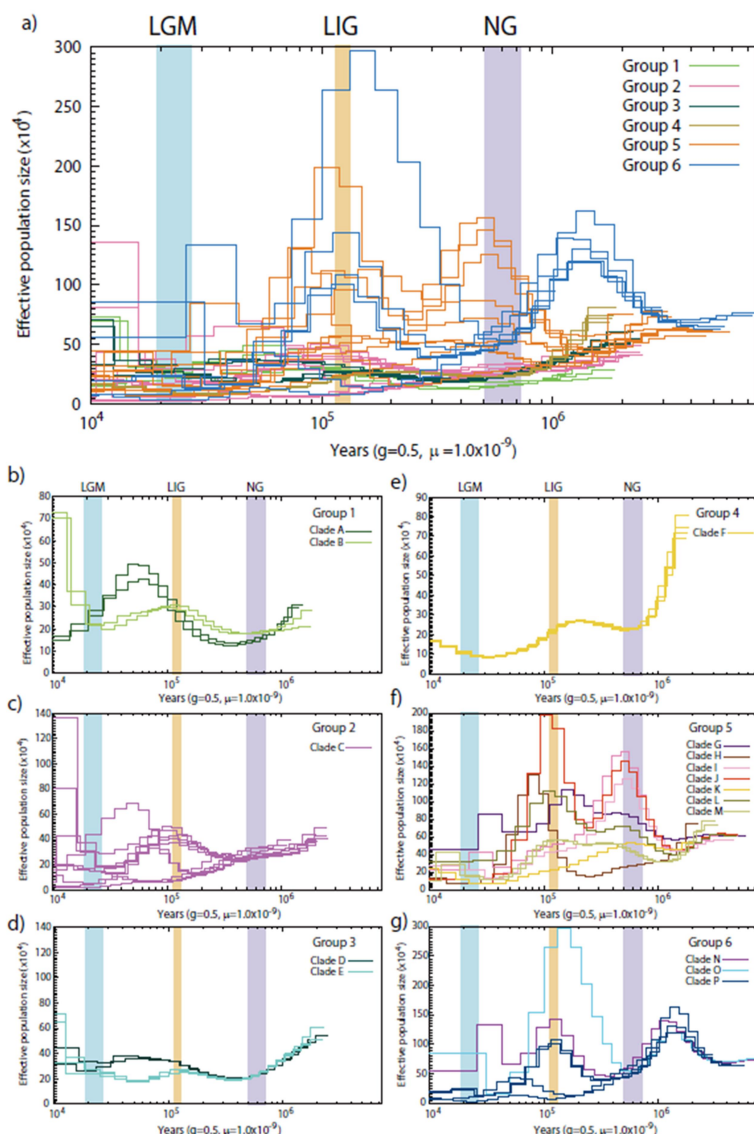


Figure 4 Demographic history of Asian shrew moles, genus *Uropsilus*: inferred changes in effective population size (N_e) over time of all six groups (a) and the clades in each group (b-g). Note the different Y-axis scale in each plot. The time scale on the X-axis is calculated assuming a neutral mutation rate (μ) = 1.0×10^{-9} per generation per site and generation time (g) = 0.5 year. The Last Glacial Maximum (LGM, 26.5 Ka – 19.0 Ka), the Last Interglacial (LIG, 129 Ka – 116 Ka), and the Naynayxungla glaciation (NG, 500 – 720 Ka) are shaded in different colors.